COGS OF AUSTRALIA FOUNDATION

Community Joint Venture Mainspring Hybrid Trust

ABN: 61 734 327 831

CORRIDORS ENERGY: ELECTRICITY

Planning and Strategy Document

Version 1.1 · CRC umbrella aligned

Issued 25 April 2026

Part of COGS Resource Corridors · Energy branch · Supersedes v1.0

A direct Member-benefit electricity supply programme delivered through a partnership with Localvolts Pty Ltd, with part-COG$ payments as a beta-test barter instrument.

Governing Law: South Australia, Australia

Drake Village, NSW 2469 · Wahlubal Country, Bundjalung Nation

PART 1 — PURPOSE AND SCOPE

1. Purpose

1.1 This document is the planning and strategy document for Corridors Energy: Electricity (previously named "COGS Community Electricity"), a Member-benefit programme of the COGS of Australia Foundation operating within the Energy branch of COGS Resource Corridors (CRC). It sets the legal, regulatory, commercial, technical, and governance shape of the programme to a level of detail sufficient for Phase 0 legal and advisory engagement, Trustee decision record execution, First Nations Advisory Council (FNAC) concept engagement, retail partner engagement with Localvolts Pty Ltd, and subsequent platform engineering hand-off.

1.2 This v1.1 document supersedes v1.0 (COGS Community Electricity Planning and Strategy v1.0) as a structural framing revision. The programme's commercial, regulatory, operational, and platform content is substantively unchanged. What has changed is the placement of this programme within the CRC umbrella: branch identification (Energy), shared-substrate reference (under the CRC Master Strategy v1.0 document rather than restated here), and cross-programme references to siblings in the Energy branch (Corridors Energy: LPG) and adjacent branches (Corridors Communications; Corridors Transport and Freight).

1.3 Corridors Energy: Electricity is designed as a direct Member benefit and a barter services beta test record for part-COG$ payments. Electricity is the highest-frequency, highest-trust recurring Member transaction the Foundation will touch. If the COG$ services-fee mechanic can clear an electricity bill reliably and lawfully, it can clear any recurring service. The programme is therefore also a deliberate proving ground for the broader COG$ barter economy.

1.4 The document is not a deed, agreement, or regulatory submission. It is an internal strategic reference that aligns cross-hub decision-making and commits the Foundation to an explicit operating posture in advance of live Member onboarding.

1A. Branch placement and umbrella reference

1A.1 Corridors Energy: Electricity sits in the Energy branch of COGS Resource Corridors, alongside Corridors Energy: LPG. The Energy branch is one of three CRC branches (Communications, Energy, Transport and Freight) established under the CRC Master Strategy and Governance Document v1.0.

1A.2 The CRC umbrella defines the common substrate on which every CRC programme operates: Member identity and KYC; COG$ tokens and the 15% services fee; Sub-Trust C and the three-tier pattern (Community Basic, Community Standard, Community Supported); the Godley ledger; evidence-vault anchoring; FNAC governance; cross-hub governance; and Phase 0 discipline. Where this document previously restated those substrate elements, it now references the CRC Master Strategy document. Substrate content is not duplicated here.

1A.3 Branch-level coordination with the Energy sibling programme: Corridors Energy: LPG operates on a different partner model (Foundation-held wholesale account with Supagas) but uses the same substrate. The two programmes share FNAC reporting, Sub-Trust C draw forecasting, and the Energy branch's GST advisory strategy, and they feed the Transport and Freight branch's LPG last-mile integration described in §22 of the CRC Master Strategy.

1A.4 Cross-branch adjacency: Corridors Communications delivers connectivity that Members use to access this programme. Corridors Transport and Freight is a downstream consumer of Energy branch

data (Localvolts pricing for EV charging optimisation is a Phase 7+ integration flagged in CRC Master Strategy §24).

2. Scope

2.1 Phase 1 of COGS Community Electricity is supply-only. The Foundation is not, and is not proposing to become, an authorised electricity retailer. The Foundation is not a generator, a meter provider, an embedded network operator, or a market participant in the National Electricity Market (NEM). The Foundation is the community-facing sponsor and AI optimisation partner sitting alongside an authorised retailer.

2.2 The Foundation's retail partner is Localvolts Pty Ltd, an AER-authorised electricity retailer operating a peer-to-peer marketplace across the Australian Capital Territory, New South Wales, Queensland, South Australia, and Tasmania. Localvolts holds the retail contract with each Member directly. The Foundation does not sit between the Member and the grid.

2.3 COGS Community Electricity is a direct Member benefit delivered under Sub-Trust C of the Community Joint Venture Mainspring Hybrid Trust structure. The programme is designed to be offered in three tiers: Community Basic, Community Standard, and Community Supported — mirroring the structure already established for COGS Community Mobile.

2.4 Geographical phasing is explicit. The Phase 1 pilot is the Jubullum, Drake, and Tabulam communities in the NSW Essential Energy distribution area. Phase 2 extends NSW-wide. Phase 3 extends to Localvolts' full NEM coverage (ACT, QLD, SA, TAS). Phase 4 addresses Victoria via a secondary retailer partner, since Localvolts is not currently authorised in Victoria.

2.5 The programme does not currently include community-scale electricity generation. Joint-venture community generation (solar, wind, or battery) is explicitly reserved for a later phase and is not within scope of this document. A Member's individual behind-the-meter solar or battery will be supported through the Localvolts peer-to-peer marketplace and the Foundation's AI optimisation layer, but the Foundation does not own, operate, or finance such installations under Phase 1.

3. Statement of alignment with the Community Mobile programme

3.1 COGS Community Electricity is structurally analogous to COGS Community Mobile. The analogy is deliberate and is designed to allow the two programmes to share Member-facing patterns, back-office accounting, governance, and AML/CTF controls.

Element	Community Mobile	Community Electricity
Member benefit type	Mobile voice and data supply	Electricity supply
Wholesale / licensed partner	Partner Wholesale Networks (PWN)	Localvolts Pty Ltd
Foundation role	Carriage Service Provider (CSP)	Community sponsor and AI optimisation partner
Tier structure	Basic / Standard / Supported	Basic / Standard / Supported
Supported-tier eligibility	First Nations, concession, Pay It Forward, state-care	First Nations, concession, Pay It Forward, state-care
COG$ payment	Class C offset up to cap	15% services fee in COG$; Class

Element	Community Mobile	Community Electricity
integration		C offset for Supported tier
Pilot geography	Drake, Tabulam, Tenterfield	Jubullum, Drake, Tabulam
First Nations governance	FNAC + JLALC engagement	FNAC + JLALC engagement
Sub-Trust funding	Sub-Trust C	Sub-Trust C

3.2 Where the two programmes diverge, this document makes that divergence explicit. The most material divergences are: (a) the Foundation is not the retail contract counterparty (Mobile has the Foundation as CSP; Electricity has Localvolts as retailer); (b) the 15% community-services fee is denominated in COG$ and characterised as a combined retail services margin and infrastructure-testing nominal value; (c) the GST treatment is flow-through via Sub-Trust C as community infrastructure support and capability-building, which changes the tax posture materially relative to Mobile; (d) the AI optimisation layer has no direct Mobile analogue and is a new technical build.

4. Document status and authority

4.1 This document is version 1.0, issued 25 April 2026. It is the founding strategic reference for the programme and is the document referenced in the COGS of Australia Foundation community projects hub (project ID to be assigned on creation) as the founding document.

4.2 Amendments to this document require a Caretaker Trustee decision record and, if the amendment affects First Nations governance or Sub-Trust C funding mechanics, FNAC concurrence before the amended version is taken as authoritative.

4.3 This document does not override the Joint Venture Participation Agreement (JVPA), the CJVM Hybrid Trust Declaration, or any Sub-Trust Deed. Where any operational choice described here would, on close reading, be inconsistent with any of those supreme instruments, the supreme instrument prevails and this document is taken to be amended to conform.

PART 2 — LEGAL AND REGULATORY POSTURE

5. Core posture statement

5.1 The Foundation's legal and regulatory posture for COGS Community Electricity is deliberately lightweight. The Foundation does not become an authorised electricity retailer, does not sit in scope of the National Electricity Retail Law (NERL) as a retail counterparty, does not become an exempt seller under Australian Energy Regulator (AER) exempt-selling arrangements, and does not become an AEMO market participant.

5.2 The Foundation's regulatory exposure is limited to four surfaces only: (a) its existing Caretaker Trustee obligations under the CJVM Hybrid Trust Declaration; (b) its ongoing Managed Investment Scheme (MIS) exclusion posture under s.9 Corporations Act 2001 (Cth) as currently managed with ASIC; (c) the AML/CTF regime as it applies to the Foundation's Member financial flows; and (d) Australian taxation law, including GST and income tax, as applied to the flow-through treatment of Sub-Trust C payments.

5.3 Every regulated activity inherent to the supply of electricity to a retail customer — retail authorisation, AEMO wholesale settlement and prudentials, network charge pass-through, renewable certificate surrender, environmental disclosure, Standing Offer and Energy Made Easy publication, Energy and Water Ombudsman membership, life-support register maintenance, consumer data right obligations, Retailer of Last Resort readiness — rests with Localvolts Pty Ltd under its existing retail authorisation.

6. NERL and AER scope analysis

6.1 Under the National Energy Retail Law, a person requires AER retailer authorisation (or a retail exemption) to sell energy to a customer at retail. The Foundation does not sell energy. The Foundation does not hold title to electricity at any point in the supply chain, does not pay AEMO for wholesale energy, does not cause the energisation or de-energisation of any Member's premises, and does not issue the retail bill. All of these actions are performed by Localvolts directly, and the Member's retail contract is a standard Localvolts Market Retail Contract to which the Foundation is not a party.

6.2 The Foundation charges a community-services fee in COG$, described in Part 3. That fee is for optimisation, community benefit administration, subsidy processing, and evidence-vault anchoring. It is not a fee for the sale or supply of electricity. The commercial characterisation of the 15% services fee as a retail services margin and infrastructure-testing nominal value is explicitly internal to the Foundation-Member community relationship; it is not a retail supply transaction and does not bring the Foundation within NERL.

6.3 The Foundation does not market electricity retail plans on behalf of Localvolts. The Foundation directs Members to Localvolts' standard onboarding pathway via a Foundation-branded co-marketing surface (“COGS Community Electricity, powered by Localvolts”). The Foundation is not an authorised representative of Localvolts and does not hold out as one. The Foundation does not discuss Localvolts' pricing, plan terms, or contract terms in its own right; those are Localvolts' representations to the Member under the retail contract.

Legal-review confirmation sought in Phase 0

Phase 0 legal review is to confirm that the arrangement described in this Part, read together with the commercial model in Part 3, does not: (a) bring the Foundation within NERL as a retailer or

marketer; (b) require authorisation or exemption under any NERL-adjacent scheme; (c) reclassify the 15% COG$ services fee as a retail supply consideration; or (d) create any other regulated-electricity-market status for the Foundation. The Foundation will not commence Member onboarding until this confirmation is on file.

7. MIS exclusion posture

7.1 The Foundation already operates an MIS exclusion posture under s.9 Corporations Act 2001 (Cth) focused on defeating the Third Limb (day-to-day control) of the MIS definition. That posture is currently active with ASIC. COGS Community Electricity is designed not to disturb that posture.

7.2 The arrangement is structured so that (a) no Member pools consideration with other Members to acquire electricity; each Member's electricity retail contract is bilateral with Localvolts; (b) no Member's financial return depends on the activities of another Member; (c) the Foundation does not undertake the essential day-to-day operations of producing or delivering electricity (Localvolts does); and (d) the 15% COG$ services fee is characterised and recorded as a services margin for community-infrastructure testing and administration, not as a contribution to a common enterprise yielding financial benefits to Members.

7.3 The programme is reviewed at Phase 0 by the Foundation's regulatory counsel to confirm consistency with the existing MIS-exclusion argument. Any adjustment required for MIS-posture integrity is made to the programme, not to the MIS posture.

8. AML/CTF

8.1 Registration with AUSTRAC in respect of the Foundation's broader activities is proceeding as required on its own timeline. COGS Community Electricity introduces a recurring AUD-and-COG$ flow between Member and Foundation that is within the Foundation's existing Member KYC perimeter. The Member has already been KYC-identified on joining the Foundation; electricity onboarding is a sub-process within that existing identity relationship.

8.2 The COG$ services-fee leg and the Sub-Trust C subsidy leg are both intra-Foundation Member transactions. They do not involve the movement of designated services under the AML/CTF Act 2006 (Cth) that would not already be in scope of the Foundation's existing controls. The Foundation's Phase 0 AML/CTF advice includes a specific check on the combined-electricity-plus-COG$ flow to confirm that the existing programme covers it.

8.3 Localvolts operates its own KYC and customer-due-diligence processes as a retailer. The Foundation does not share Member KYC data with Localvolts beyond what is required for retail contract onboarding with the Member's informed consent (name, address, ID documents, life-support declarations where applicable).

9. Energy and Water Ombudsman and consumer-protection posture

9.1 The Member's retail-service complaints pathway is through Localvolts and, where Localvolts does not resolve a complaint, through the Energy and Water Ombudsman of New South Wales (EWON) or the equivalent body in other jurisdictions. Localvolts is responsible for its EWON membership and complaints responsiveness.

9.2 The Foundation maintains an internal Member support channel for community-services issues (COG$ invoicing, Sub-Trust C subsidy application, AI optimisation guidance), which is separate from

the electricity retail complaints channel. Where a Member raises an issue that is in scope of Localvolts' retail obligations, the Foundation directs the Member to Localvolts and offers to assist in escalating through EWON if required.

9.3 The Foundation does not offer advice that could be construed as financial product advice, tax advice, or legal advice to Members in respect of their Localvolts retail contract. Foundation communications are Foundation-services-only.

10. Life-support obligations

10.1 Life-support obligations under NERL are Localvolts' responsibility as the retailer. The Foundation's Member onboarding surface includes an explicit prompt for Members to declare life-support equipment on the premises so that Localvolts is correctly informed and the life-support register is accurately maintained from Day 1.

10.2 The Foundation's AI optimisation layer does not issue device-control signals that could interfere with life-support equipment. The optimisation layer is advisory in Phase 1 and in no case overrides any safety-critical equipment.

11. Retailer of Last Resort (RoLR) continuity

11.1 If Localvolts is ever subject to a RoLR event, Members are automatically transferred to the default RoLR for their distribution area under the AER scheme. Members experience no loss of electricity supply. The Foundation-Member COG$ relationship continues as a separate services relationship that is not affected by the RoLR event; however, the Foundation would pause 15% services-fee invoicing until the Member's new retail contract is clarified.

11.2 In the event of a RoLR event, the Foundation would promptly seek a new retail partner arrangement. Candidates already surveyed in Phase 0 planning include Amber Electric, Next Business Energy Pty Ltd (which holds the Indigo Power Retail licence), and Diamond Energy. Rebranding onto a replacement retail host is a planned contingency.

12. Data protection

12.1 The Foundation's AI optimisation layer reads NEM-period electricity data for consenting Members via the Localvolts Partner API. This data is de-identified at rest where practicable and is held under the Foundation's existing data-protection governance (evidence vault anchoring, access-control, and data-minimisation rules).

12.2 Member consent for Foundation API read access is obtained explicitly during COGS Community Electricity onboarding as a separate consent to the retail contract with Localvolts. Consent is revocable; revocation removes the Member from the AI optimisation layer and reduces the 15% services-fee line to the base services component.

12.3 The Foundation is subject to the Privacy Act 1988 (Cth) via Australian Privacy Principles where applicable to its operations. Electricity-optimisation data handling is added as a new section to the Foundation's existing privacy notice during Phase 2.

PART 3 — COMMERCIAL MODEL

13. Plain-language description

13.1 The commercial model has two independent legs. The first leg is a standard retail electricity supply relationship between the Member and Localvolts, priced entirely in AUD, under Localvolts' existing Market Retail Contract. The Foundation is not a party to this leg. The second leg is a community services agreement between the Member and the Foundation, priced in COG$, calculated as 15% of the Member's AUD Localvolts bill for the billing period. These two legs operate in parallel and are kept structurally separate.

13.2 For the Community Supported tier, the Foundation pays the Localvolts AUD bill in full from Sub-Trust C, and the Member's 15% COG$ community services fee is either drawn from the Member's Class C COG$ balance (where available) or waived for the period (where the Member's balance is insufficient and they fall within Sub-Trust C eligibility). The Member experiences the programme as zero or near-zero net cost.

14. The three tiers

Tier	Member experience	Funding and constraints
Community Basic	Member pays Localvolts AUD bill in full; pays 15% COG$ services fee to Foundation; access to daily AI optimisation guidance in the Member dashboard.	No Sub-Trust C draw. Modest Class C COG$ offset permitted against the services fee line, subject to per-Member monthly and annual caps.
Community Standard	Member pays Localvolts AUD bill in full; pays 15% COG$ services fee; access to 5-minute real-time AI optimisation signals for compatible smart devices.	No Sub-Trust C draw. Larger Class C COG$ offset available against the services fee line, subject to caps.
Community Supported	Sub-Trust C pays Localvolts AUD bill in full; Member's 15% COG$ services fee is drawn from Class C balance or waived; full access to optimisation guidance.	Sub-Trust C fully funds the AUD leg. Eligible Members: First Nations Members; concession card holders; Pay It Forward recipients; persons under state care.

14.1 Tier assignment is determined at onboarding and reviewed annually. Movement between tiers is permitted; movement into Community Supported requires eligibility verification on the Foundation side.

14.2 The Board will approve the specific numeric values for per-Member Class C COG$ offset caps (monthly and rolling 12-month) at Phase 3 of the build. Electricity bills can be large and highly variable month-to-month relative to mobile, and the cap is an explicit risk control. The default design assumption is a cap that protects Members from unintentionally over-burning their Class C balance during a summer spike, and protects the Foundation from absorbing runaway wholesale-price events through Sub-Trust C.

15. The 15% community services fee — characterisation

15.1 The 15% fee is characterised as a combined retail services margin and infrastructure-testing nominal value. This characterisation has three operational consequences.

15.2 First, the fee is for Foundation-delivered services: AI optimisation read-and-advise layer, Member dashboard, Sub-Trust C subsidy administration, community support channel, evidence-vault anchoring of settlement events, and compliance with the Foundation's Godley accounting discipline. These are bona fide services; the fee is bona fide consideration.

15.3 Second, the fee is a nominal value used to test the COG$ barter-payment mechanic. Electricity is the highest-frequency, highest-trust recurring Member transaction the Foundation will touch. Observing how the COG$ services-fee invoicing, settlement, caps, and waivers perform in a live electricity-billing context provides evidence for the broader COG$ economy across every subsequent service. A low, predictable, percentage-based fee is well suited to producing that evidence without distorting Member behaviour.

15.4 Third, the fee is denominated in the Foundation's own internal community token (COG$) and is internal to the Foundation-Member community relationship. It is not consideration for supply of electricity; it is consideration for community services delivered by the Foundation to the Member.

16. GST treatment

16.1 The Foundation's position, confirmed in Phase 0 ATO advice or private ruling, is that GST may not be chargeable on the 15% COG$ services fee where the transaction is structured to flow through Sub-Trust C as community infrastructure support and capability-building payments. The flow-through treatment rests on three characterisations: (a) COG$ itself is not Australian currency and is not a digital currency that has been declared equivalent for GST purposes under A New Tax System (Goods and Services Tax) Act 1999 (Cth); (b) the Foundation is operating Sub-Trust C for community-infrastructure and capability-building purposes on behalf of the community of Members; and (c) the 15% services fee's characterisation as a nominal infrastructure-testing value is consistent with non-monetary consideration used for internal community purposes.

16.2 The Foundation does not rely on the informal description above in production. Phase 0 Item 6 (see Part 6) is to obtain ATO-confirmed guidance, including if appropriate a private ruling request, on the GST treatment of the 15% COG$ services fee and the Sub-Trust C subsidy. The programme's operational GST posture is set by whatever the ATO advice returns, not by this document.

16.3 GST on the AUD leg (Member to Localvolts) is unaffected by this analysis. Localvolts issues a standard GST-compliant tax invoice to the Member as part of its ordinary retail billing. For Community Supported tier Members, the Foundation's payment to Localvolts from Sub-Trust C is made in AUD and Localvolts issues its invoice to the Member (or to the Foundation on the Member's behalf) on its normal GST terms. GST on the AUD leg is not disturbed.

Flow-through framing for Sub-Trust C

Sub-Trust C is the Foundation's community benefit sub-trust. Its operational purpose explicitly includes community infrastructure support and capability-building. Directing the 15% COG$ services-fee flow through Sub-Trust C as a community-infrastructure-and-capability payment establishes the Sub-Trust C purpose from the first transaction. The flow-through treatment is therefore not a tax-avoidance posture; it is the plain-English description of what is actually happening in the ledger.

17. Worked example

17.1 The following worked example illustrates the commercial flow for each tier using a 90-day billing period and an indicative AUD Localvolts bill total of $450.

Flow	Basic	Standard	Supported
Member to Localvolts (AUD)	$450.00	$450.00	$0.00
Sub-Trust C to Localvolts (AUD)	$0.00	$0.00	$450.00
Foundation services fee (COG$ value equivalent)	67.50 COG$	67.50 COG$	67.50 COG$
Member to Foundation (COG$ cash)	67.50 COG$	67.50 COG$	0 COG$
Class C offset available (up to cap)	Small	Larger	Up to full fee
Net Member out-of-pocket (AUD equivalent)	~$450 + services fee value	~$450 + smaller net services fee	~$0

17.2 The 15% fee produces a COG$ flow to the Foundation that scales naturally with Member electricity consumption, without any incentive for the Foundation to inflate consumption. The Foundation's incentive sits with the AI optimisation layer: a well-optimised Member has a lower AUD bill, a proportionally lower 15% fee, and a happier community relationship.

18. Commercial partnership with Localvolts

18.1 The Foundation's commercial ask of Localvolts is modest and preserves Localvolts' existing economics. The ask is set out in §30 of the Concept Note issued to Localvolts and summarised here for internal reference: (a) a business/partnership account structure that preserves individual Member retail contracts with Localvolts; (b) a Partner API key and Partner ID for the AI optimisation layer with NMI wildcard access across the consenting Member fleet; (c) a co-branded onboarding path; and (d) an optional Foundation-referral acknowledgement (not required by the Foundation; any fee offered flows to Sub-Trust C, not to Foundation operating revenue).

18.2 The Foundation will not pay Localvolts any aggregation fee, platform fee, or channel management fee. The Member pays Localvolts on standard retail terms. The Foundation's cost base for the partnership is internal (build and operation of the AI optimisation layer, Member dashboard, and subsidy administration), not paid to Localvolts.

18.3 If Localvolts declines the partnership or offers terms the Foundation is unable to accept, the Foundation's fallback is a white-label arrangement with Next Business Energy Pty Ltd (which holds the Indigo Power Retail licence and is the closest equivalent community-energy white-label host in Australia), with Amber Electric as a parallel second-offer option for Members with batteries and EVs. These fallback options are held in reserve and are not actively pursued in Phase 1.

19. Pricing sensitivity

19.1 At 1,000 active Members with an average AUD bill of $1,600 per year, the 15% services-fee flow produces approximately $240,000 per year in COG$ services-fee income at face value, with 30% minimum allocated to First Nations community benefits under the entrenched Sub-Trust C provision. At 10,000 Members the flow is approximately $2.4 million per year at face value.

19.2 The 15% figure is proposed as the Phase 1 starting point. It is reviewed at 6 months, 12 months, and 24 months against Member tier distribution, Class C COG$ offset usage, Sub-Trust C draw accuracy, and general Member affordability. The Board retains the power to adjust the percentage by decision record, subject to FNAC concurrence where the change affects the Community Supported tier.

PART 4 — LOCALVOLTS PARTNERSHIP

20. Why Localvolts is the Phase 1 partner

20.1 Four structural features make Localvolts the natural Phase 1 retail partner:

- **Zero-markup pass-through pricing.** Localvolts' public position that all prices, costs and charges are applied on a pass-through basis with no usage markup is structurally identical to the commercial posture the Foundation wants to offer its Members. No other AER-authorised retailer in the NEM offers this.
- **Partner API built for exactly this use case.** The Localvolts API (v0.9.2) exposes 5-minute NEM-period pricing, flex-up and flex-down fields, CO2e content, and quality-flagged forecasts. It is authentication-secured via API key plus Partner ID in the HTTP header, designed explicitly for external industry-partner integration.
- **Coverage matches the pilot geography.** Localvolts is AER-authorised in ACT, NSW, QLD, SA, and TAS. All three pilot communities (Jubullum, Drake, Tabulam) are in NSW Essential Energy territory. Phase 1 is covered from day one; Phases 2 and 3 are covered without adding a partner.
- **Values alignment.** Localvolts' public materials describe an ambition to support community groups and First Nations engagement. A First Nations-partnered community joint venture running an AI optimisation layer on the Localvolts API is a natural showcase partnership, not a commodity retail deal.

21. Partnership structure

21.1 The partnership is a simple services alignment, not a joint venture, authorised-representative relationship, or distribution agreement. The shape is:

- **Member ↔ Localvolts:** standard Localvolts Market Retail Contract, unchanged from Localvolts' public offering. Localvolts handles onboarding, billing, support, complaints, EWON, life-support, and all retail obligations.
- **Foundation ↔ Localvolts:** a simple written Partnership Arrangement covering (a) Partner API access scope and security, (b) onboarding-channel acknowledgement, (c) any co-branding parameters Localvolts is comfortable with, and (d) optional Foundation-referral acknowledgement. This is a short document, probably one to three pages.
- **Member ↔ Foundation:** a community services terms-of-use executed at COGS Community Electricity onboarding. Covers the 15% COG$ services fee, AI optimisation layer consent, Sub-Trust C eligibility framework, data-handling, and Class C offset rules.

22. Localvolts API integration — technical scope

22.1 The Foundation's Partner API integration consumes the Localvolts production API at api.localvolts.com under v1. Authentication uses the Foundation's issued API Key and Partner ID in HTTP headers as described in the Localvolts API Guide v0.9.2, section 1.1.

22.2 Endpoints used in Phase 1 (read only):

Endpoint	Purpose	Notes
GET	Pull current and forward NEM-	Default returns current 5-min

Endpoint	Purpose	Notes
/v1/customer/interval?NMI=* (default window)	period data across the consenting Member fleet.	period plus all periods out to 24 hours.
GET /v1/customer/interval?NMI=*&from=...&to=...	Pull a historical 24-hour window for post-event reconciliation and Godley ledger anchoring.	Historical range limited to 72 hours in the past; one-call window is 24 hours maximum.
GET /v1/version	Periodic version-probe for the API. No auth required.	Foundation monitors API version to detect breaking changes early.

22.3 Phase 2 endpoints (when Localvolts releases POST /customer/interval in production): the Foundation becomes a Day 1 adopter of the device-submission endpoint for Members who opt into closed-loop optimisation (battery SOC feedback, appliance state reporting). Device-submission is not used in Phase 1.

22.4 API call pattern: the Foundation polls /customer/interval every 5 minutes during active optimisation hours and every 30 minutes at other times, plus an end-of-day sweep pulling actuals for the completed trading day. All response payloads are anchored to the evidence vault as settlement events (see §45).

23. Data fields and their use

23.1 The Localvolts customer-interval response provides the fields used by the Foundation's AI optimisation layer and Member dashboard:

Field	Role in the optimisation layer
intervalEnd (UTC)	Canonical timestamp for every NEM period used in anchoring and evidence.
exportsAll / importsAll	Volume signals used for tier consumption reporting and monthly reconciliation.
earningsAllVarRate / costsAllVarRate	Headline c/kWh rate seen by the Member for the period, driving “current price” display.
earningsFlexUp / costsFlexUp	The marginal export/import price change for an additional kWh. Primary signals for the real-time 5-minute optimiser.
earningsFlexDown / costsFlexDown	The marginal export/import price change for a reduced kWh. Used to decide whether shifting load is worthwhile.
exportsAllEmissions / importsAllEmissions	Grams CO2e content per kWh. Feeds the Foundation's carbon-aware optimiser and Member environmental dashboard.
quality (Act / Exp / Fcst)	Confidence weight for the period data. Optimiser weights Act = 1.0, Exp = 0.7, Fcst = 0.4 in the control loop.
lastUpdate (UTC)	Freshness indicator; optimiser ignores periods older than 60

Field	Role in the optimisation layer
	seconds during active windows.

24. Co-branding and onboarding

24.1 The Foundation-branded onboarding pathway presents the programme as "COGS Community Electricity, powered by Localvolts." The Member experiences:

- Foundation Member login and tier selection.
- Foundation eligibility check (for Community Supported tier only).
- Foundation-side data-consent capture (API optimisation read access; Sub-Trust C subsidy consent where applicable).
- Handoff to Localvolts' standard onboarding for retail contract, identity verification, NMI lookup, and life-support declaration.
- Return to the Foundation dashboard with a confirmation that the retail contract is live.

24.2 The Member experiences one coherent journey but is always shown clearly that (a) their retail contract is with Localvolts, (b) the community services relationship is with the Foundation, and (c) the two are commercially independent.

25. Engagement sequence

25.1 Initial engagement with Localvolts follows this sequence:

- **Step 1.** Foundation issues the Localvolts Partnership Concept Note v1.0 (this document's companion) to the named points of contact. Expected response window: two to four weeks.
- **Step 2.** Exploratory call between the Foundation's Caretaker Trustee and Localvolts' founder or partnerships team.
- **Step 3.** If mutual interest, Foundation prepares a short draft Partnership Arrangement; Localvolts reviews.
- **Step 4.** Partnership Arrangement executed; Partner API credentials issued.
- **Step 5.** Technical integration begins under the Phase 2 build (see Part 8).

25.2 If Localvolts declines or is slow to engage, the Foundation proceeds to the fallback described in §18.3 without adjusting the rest of the programme architecture. The design is retailer-partner-portable.

26. Fallback retailer candidates (not Phase 1)

26.1 Fallback candidates are held in reserve and are not actively engaged in Phase 1. They are listed here for contingency completeness:

Candidate	Fit	Relative position
Next Business Energy Pty Ltd	Holds the licence under which Indigo Power Retail operates; proven community-energy white-label host.	Primary fallback. Closest analogue to PWN on the Mobile side.
Amber Electric	Monthly subscription + true	Secondary fallback, or parallel

Candidate	Fit	Relative position
	wholesale pass-through; strong API; values-aligned.	offer for Members with batteries and EVs.
Diamond Energy	Wholesale hedge counterparty to prior community retailers; supplies small retailers.	Third-option fallback, Victoria-capable for Phase 4.

PART 5 — AI OPTIMISATION ARCHITECTURE

27. Purpose of the optimisation layer

27.1 The Foundation's AI optimisation layer sits above Localvolts' peer-to-peer marketplace and API. Its purpose is to (a) deliver tangible value to Members beyond what a typical retail plan provides, (b) convert the Localvolts real-time NEM data into Member-actionable advice, (c) aggregate Member flexibility into signals that inform the Foundation's broader Sub-Trust C forecasting, and (d) produce high-quality evidence for the COG$ barter-services beta test.

27.2 The layer is advisory in Phase 1. It does not directly control Member appliances, batteries, or EVs. Device control is a Phase 3 consideration subject to safety review, life-support carve-outs, and explicit per-device Member consent.

28. Three optimisation loops

28.1 The optimisation layer operates at three distinct timescales, each serving a different Member decision.

28.2 Fast loop (5-minute real-time)

- Input: latest /customer/interval response for the Member's NMI.
- Use: advise Members with compatible devices when to charge or discharge a battery, when to run or defer high-draw appliances, when to export or self-consume solar.
- Output: push-notification or dashboard indicator — “green now”, “amber now”, “red now” — based on costsFlex and emissions signals, weighted by the Localvolts quality flag.
- Members without smart devices receive this loop as a simple live dashboard indicator only.

28.3 Daily loop (24-hour planning)

- Input: 24-hour forward window of /customer/interval responses, combined with the Member's historical consumption shape.
- Use: plan EV charging windows, pool pumps, hot water, dishwashers and washing machines; flag likely negative-price windows where the Member earns by consuming.
- Output: overnight email or app summary the night before — “tomorrow's cheapest 3-hour block is 11:00–14:00; schedule the EV for that window.”

28.4 Seasonal / model loop (monthly and quarterly)

- Input: the Member's historical /customer/interval data over the last 30, 90 and 365 days, aggregated across the Member fleet where relevant.
- Use: forecast the Member's next bill, recommend whether a battery or EV would pay back, flag Members trending toward Class C offset exhaustion, budget Sub-Trust C draw for the Community Supported tier.
- Output: monthly Member report plus Foundation-internal Sub-Trust C draw forecast used in Godley cash planning.

29. Model and training approach

29.1 The Phase 1 optimiser is a rule-based layer, not a machine-learning model. The ruleset is derived directly from the Localvolts field semantics (“signal that costsFlexUp is above X means defer load”) and

is deterministic, auditable, and easy to explain to Members and regulators. Rule parameters (thresholds, windows) are Board-approved and tracked in the Foundation's configuration register.

29.2 Phase 2 introduces a statistical load-profile model per Member (simple time-of-day and weather-conditional usage forecast) that improves the daily-loop accuracy. No Member-private data leaves the Foundation's infrastructure for training; the model is trained on the Member's own historical data held under the Foundation's existing data-governance framework.

29.3 Phase 3 and beyond contemplate a fleet-level optimisation model that aggregates Member flexibility (where consented) to offer virtual-power-plant-style services into demand-response or frequency-control markets. This is not within Phase 1 scope and is subject to its own separate planning document.

30. Technical architecture overview

30.1 The Foundation's optimiser is a server-side service running on the existing COGS platform infrastructure (PHP 8.4 / MariaDB 10.6 / Apache / cPanel Serversaurus). The architecture has four components:

- **LocalvoltsApiClient (service).** Handles authentication, rate-limiting, retries, and parsing of the Localvolts /customer/interval JSON payload. Single point of integration with the upstream API.
- **ElectricityIntervalStore (database).** MariaDB table holding normalised NEM-period records per NMI per interval, with quality flags and lastUpdate timestamps. Source of truth for the Member dashboard and the Godley anchoring.
- **OptimiserEngine (service).** Rule-based engine reading from ElectricityIntervalStore, producing per-Member advice records and fleet-level summaries. Runs on a 5-minute cron for the fast loop, an overnight cron for the daily loop, and a monthly cron for the seasonal loop.
- **MemberDashboard (UI).** /services/electricity/ pages in the Member Independence Vault area. Live tile for the fast loop, tomorrow view for the daily loop, monthly view for the seasonal loop.

31. Data-protection controls specific to the optimiser

31.1 Member NMI data is treated as sensitive. Access is restricted to the LocalvoltsApiClient and ElectricityIntervalStore services. The Member dashboard displays only the Member's own data. Admin access is restricted to Foundation staff under the existing admin access-control model.

31.2 Data minimisation: only fields actively used by the optimiser or the Member dashboard are retained. Full raw payloads are retained for 30 days for reconciliation, then rolled into summary records. The raw payload is anchored to the evidence vault at ingestion to preserve audit integrity without retaining full payload indefinitely.

31.3 Consent revocation: a Member revoking API-read consent triggers (a) cessation of polling for that Member's NMI, (b) removal of stored interval records older than the minimum retention period required for billing reconciliation, and (c) a tier downgrade from Standard to Basic (since the optimisation layer content the Standard tier offers depends on API-read access).

32. Evidence-vault anchoring for the optimisation layer

32.1 The optimisation layer's interaction with Member data is anchored to the Foundation's evidence vault at two points: (a) each API poll response batch is hashed and anchored, producing an auditable record that the data received on the Foundation side is identical to the data Localvolts emitted; (b) each

per-Member billing-period reconciliation (Member bill vs Foundation-held data) is anchored as a reconciliation record.

32.2 Anchoring uses the Foundation's existing SHA-256 hashing and evidence-vault discipline. No Member-identifiable content is published on-chain; only content hashes. This preserves Member privacy while providing cryptographic integrity for audit and regulatory inspection.

33. Risk controls in the advisory layer

33.1 The optimiser does not override safety-critical equipment. Life-support-declared premises are flagged in the ElectricityIntervalStore and are excluded from any advice that would recommend reducing or interrupting supply. The advisory layer is also configured to err on the side of supply-preserving advice in all edge cases.

33.2 Advice quality is measured. Each piece of advice is logged with its input conditions and, where measurable, its realised outcome. Drift in advice quality (for example, signal-based “green now” recommendations that routinely miss the actual cheapest window) triggers an engineering review before the next ruleset update.

PART 6 — PHASE 0 READINESS

34. Phase 0 purpose and scope

34.1 Phase 0 is the pre-launch readiness phase. Nothing in Phase 1 proceeds until Phase 0 is complete and the Caretaker Trustee records a decision confirming Phase 0 closure. Phase 0 is not about customer acquisition or platform build; it is about establishing the legal, regulatory, governance, and advisory foundations on which the rest of the programme can safely stand.

34.2 Phase 0 is targeted to complete in the May–July 2026 window, ahead of Localvolts commercial engagement and independent of Governance Foundation Day (14 May 2026). The Community Mobile programme's Phase 0 cycle is the template; this programme's Phase 0 adapts that template for electricity-specific items.

35. Phase 0 checklist

35.1 Phase 0 consists of ten numbered items. Each item has a Responsible owner, a Success condition, and a Decision-record trigger on completion.

#	Item	Owner	Success condition
1	Trustee decision record authorising COGS Community Electricity scope	Caretaker Trustee	Signed TDR filed; decision_category populated; document published internally
2	FNAC concept engagement; determination whether Jubullum is in the first cohort or reviews after Drake/Tabulam proof	Caretaker Trustee, FNAC liaison	FNAC written response received; JLALC (Ken Avery, Priscilla Bell) formally briefed; cohort composition confirmed
3	GST and tax advice on 15% COG$ services fee and Sub-Trust C flow-through	Foundation's tax adviser	Written advice received; (if required) private-ruling application lodged with ATO
4	Legal review of characterisation of the 15% fee and of the NERL/MIS posture	Foundation's regulatory counsel	Written legal opinion on file confirming NERL non-scope, MIS-posture consistency, and fee characterisation
5	AML/CTF advice refresh covering the combined electricity-plus-COG$ flow	AML/CTF adviser	Written confirmation on file that combined flow is within existing Member KYC perimeter
6	Distributor ground-truth check (NMI, meter type, solar export, controlled load) across Drake and Tabulam	Foundation operations	Spreadsheet on file identifying meter types, smart-meter coverage, solar export capacity for

#	Item	Owner	Success condition
			each consenting pilot premise
7	Community information briefing(s) at Drake, Tabulam, and Jubullum	Caretaker Trustee, JLALC liaison	Attendance recorded; questions logged; any dissent noted and resolved
8	Localvolts Partnership Concept Note issued and exploratory call completed	Caretaker Trustee	Call notes filed; Localvolts response (interest or decline) recorded; fallback decision triggered only if declined
9	Life-support-register handover protocol agreed between Foundation and Localvolts	Foundation operations + Localvolts ops	Written protocol on file covering intake, escalation, and annual confirmation
10	Phase 0 closure decision record	Caretaker Trustee	Signed TDR referencing items 1–9 complete; authorising Phase 1 commencement

35.2 Items 3 and 4 (GST/tax and legal review) are on the critical path. Items 6 and 7 (distributor ground-truth and community briefings) can run in parallel with items 3 and 4 but do not gate Phase 1 completion. Item 8 (Localvolts engagement) can begin once item 1 is signed and does not need to wait for items 3–5.

36. Expected deliverables from Phase 0

36.1 At the end of Phase 0 the Foundation has on file:

- A signed Trustee Decision Record authorising the programme scope and commencement (Item 1).
- An FNAC written response confirming pilot-cohort composition (Item 2).
- Written tax advice or ATO private-ruling acknowledgement on GST treatment (Item 3).
- Written legal opinion on NERL non-scope, MIS-posture consistency, and fee characterisation (Item 4).
- Written AML/CTF confirmation on the combined flow (Item 5).
- NMI and meter inventory spreadsheet for the pilot cohort (Item 6).
- Community information briefing attendance and Q&A log (Item 7).
- Localvolts response to the concept note, with either (a) a draft Partnership Arrangement progressing, or (b) a documented decline that triggers fallback (Item 8).
- Written life-support-register handover protocol (Item 9).
- Signed Phase 0 closure Trustee Decision Record (Item 10).

PART 7 — PILOT DESIGN AND MEMBER ONBOARDING

37. Pilot cohort

37.1 The Phase 1 soft launch is a closed cohort of 30 to 60 Member premises across three communities: Jubullum, Drake, and Tabulam. All three are in the NSW Essential Energy distribution area, which allows a single distributor engagement path and a single jurisdictional overlay.

37.2 Target cohort composition (subject to FNAC concurrence under Phase 0 Item 2):

Community	Target premises	Notes
Jubullum	5–15	All on JLALC land or in close proximity. Supported tier preference for First Nations Members under the entrenched 30% Sub-Trust C allocation.
Drake	10–25	Includes the Drake Village Resource Centre and adjoining residential premises. Foundation coordinator's home community.
Tabulam	10–20	Includes the Tabulam community centre premises and surrounding residential Members.
Total	30–60	Closed cohort for a six-month beta; general availability opens at Phase 5.

37.3 The cohort is deliberately small enough that any operational issue (billing reconciliation, Sub-Trust C draw timing, COG$ invoicing edge cases) is tractable and transparent to the Foundation, and large enough to produce meaningful variance in billing shape, usage profiles, and tier distribution.

38. Onboarding flow — Member experience

38.1 The Member onboarding flow is presented within the Foundation's existing Member Independence Vault (IV) area at /services/electricity/. The Member experiences a single coherent journey that transitions seamlessly to Localvolts for the retail contract and back to the Foundation for the community services terms.

38.2 Nine-step flow:

Step	Stage	Description
1	Eligibility check	Foundation determines Member's eligible tier (Basic/Standard/Supported) from Member profile data. Supported tier triggers an explicit additional verification step.
2	Tier selection and disclosure	Member confirms tier and reads combined disclosure: Localvolts retail contract, 15% Foundation services fee in COG$, Class C offset and caps, Sub-Trust C subsidy terms

Step	Stage	Description
		where applicable.
3	NMI and address capture	Member provides NMI (if known) and confirms service address. Foundation checks distribution-area eligibility.
4	Life-support declaration	Explicit life-support declaration prompt. Any life-support flag is captured and carried into the Localvolts onboarding.
5	API-read consent	Member consents to Foundation API-read of their Localvolts interval data for AI optimisation. Consent is explicit, timestamped, and revocable at any time.
6	Handoff to Localvolts	Browser redirect to Localvolts onboarding with pre-populated fields from the Foundation flow (where Localvolts API permits).
7	Localvolts retail contract	Member executes the Localvolts Market Retail Contract under Localvolts' own onboarding process. Foundation does not see or handle contract terms.
8	Return to Foundation	Return to Foundation dashboard; Foundation records the Localvolts account identifier for API read access and services-fee invoicing alignment.
9	Welcome and first optimisation	Member receives welcome email; first AI optimisation update is generated within 24 hours after first full interval-data cycle is captured.

38.3 The nine-step flow is built in /services/electricity/onboard.php with supporting PHP services and evidence-vault anchoring of each consent event.

39. First Nations engagement protocol

39.1 Jubullum-area premises are onboarded under an explicit First Nations engagement protocol that includes: (a) direct FNAC oversight of the cohort composition from the Jubullum community, (b) JLALC briefing at each stage (concept, onboarding ready, soft launch live, 3-month review, 6-month review), and (c) a Wahlubal Country-relevant welcome and information session before any Member commits to onboarding.

39.2 The Foundation will not pressure any Jubullum community member to join the pilot. If the FNAC or JLALC determines at any point that the programme is not yet ready for Jubullum participation, that determination is respected and Drake and Tabulam proceed alone while Jubullum reviews. First Nations participation is an act of partnership, not of customer acquisition.

40. Success conditions for the pilot

40.1 The pilot is judged successful against five conditions:

- **Operational integrity.** All Localvolts bills are correctly issued to all cohort Members; all 15% COG$ services-fee invoices are correctly calculated; all Sub-Trust C subsidy payments settle within the 2-business-day transfer deadline under SubTrust C clause 7.4.

- **AI optimisation usefulness.** Member-reported usefulness of the optimisation layer (via a simple quarterly survey) is at least 3 out of 5 on average, with no individual Member reporting the layer as actively harmful.
- **Class C offset behaviour.** No Member's Class C COG$ balance is unintentionally exhausted; no Member reports the offset caps as restrictive in a way that degrades the programme's intent.
- **First Nations governance integrity.** FNAC reports the programme as operating consistent with its endorsement; JLALC reports no issues with the Jubullum-community experience.
- **Localvolts partnership health.** Localvolts reports the partnership as working commercially and operationally; no retailer-partner friction.

40.2 Pilot review is formal at 3 months and 6 months. The 6-month review triggers either Phase 5 (general availability) or a further 3-month remediation phase. Both paths are planned for; the decision is data-driven, not schedule-driven.

PART 8 — PLATFORM BUILD

41. Overview

41.1 The platform build extends the existing COGS platform (PHP 8.4 / MariaDB 10.6 / Apache / cPanel Serversaurus). It does not introduce new technology stacks. Electricity is delivered as a new feature surface, not a new platform.

41.2 The build follows the deploy discipline used throughout the Foundation's platform: SQL → PHP → HTML/JS; ground-truth check before every commit; explicit staging by filepath; PHP lint and structure checks before every package. This discipline is already established and is inherited for this build.

42. Module structure

42.1 The module split follows the existing Community Mobile pattern exactly. Two top-level directories are added:

Path	Purpose
/services/electricity/	Member-facing module. Onboarding, tier management, consent capture, dashboard, bill view, optimisation advice display.
/admin/electricity/	Back-office module. Member cohort management, Localvolts API key admin, subsidy processing, Sub-Trust C draw forecasting, support case management.

42.2 /services/electricity/ — Member pages

- **index.php** — landing page showing current tier, latest bill summary, latest optimisation advice, quick access to onboarding if not yet active.
- **onboard.php** — nine-step onboarding flow (see §38).
- **tier.php** — tier view and change request (with supported-tier eligibility check where applicable).
- **dashboard.php** — live optimisation dashboard. Fast-loop indicator, daily-loop overnight plan, seasonal view.
- **bills.php** — view of Localvolts bills summary (amount, period, reconciled status), 15% COG$ services-fee invoices issued by the Foundation, Class C offset history, Sub-Trust C subsidy history.
- **consent.php** — data-consent review and revocation.

42.3 /admin/electricity/ — back-office pages

- **cohort.php** — Member cohort list with tier, NMI, distributor, onboarding status, life-support flag, and last bill summary. Filter and search.
- **api_credentials.php** — Localvolts Partner API key management (with secrets held server-side, never exposed to the client).
- **intervals.php** — raw /customer/interval data inspector for operations and support.

- **billing_reconciliation.php** — reconciliation between Localvolts-reported usage and Foundation-side interval data; flags discrepancies.
- **subsidy_processing.php** — Sub-Trust C subsidy batch processing for the Community Supported tier.
- **services_fee_run.php** — monthly 15% COG$ services fee run (preview and commit).
- **draw_forecast.php** — fleet-level Sub-Trust C draw forecast produced by the seasonal optimiser loop.
- **cases.php** — Member support case log.

43. Database schema additions

43.1 The build adds a small, disciplined set of MariaDB tables, all prefixed electricity_ for clarity. Monetary values follow the Foundation's existing convention: integer cents for whole-dollar amounts, DECIMAL(12,4) for per-unit prices (consistent with the asx_trades convention).

Table	Purpose
electricity_member_accounts	One row per Member participating in the programme. Columns: member_id, tier, localvolts_customer_ref, nmi, distributor, onboarded_at, life_support_flag, api_consent_granted_at, api_consent_revoked_at.
electricity_intervals	Normalised /customer/interval records. Composite key (nmi, interval_end_utc). Columns include imports_wh, exports_wh, cost_var_rate_c_per_kwh DECIMAL(12,4), earnings_var_rate_c_per_kwh, flex_up_c, flex_down_c, emissions_g_co2e, quality ENUM('Act','Exp','Fcst'), last_update_utc, raw_payload_sha256.
electricity_bills	One row per Localvolts billing period per Member. Columns: member_id, period_start, period_end, total_aud_cents, localvolts_invoice_ref, reconciled_at, reconciliation_status.
electricity_services_fees	Foundation's 15% COG$ invoices. Columns: member_id, bill_id, fee_cogs_cents, issued_at, settled_at, class_c_offset_applied_cogs_cents, waiver_reason.
electricity_subsidies	Sub-Trust C AUD payments made on behalf of Supported-tier Members. Columns: member_id, bill_id, aud_cents, stc_transfer_ref, transferred_at.
electricity_optimiser_advice	Log of optimisation advice issued per Member. Columns: member_id, loop (fast/daily/seasonal), input_window_start, input_window_end, advice_body, quality_weight, issued_at.
electricity_support_cases	Member-facing support case log. Foundation-side only; Localvolts retail complaints are handled separately via Localvolts' own systems.

44. Integration with existing platform services

44.1 The module integrates with existing platform services rather than duplicating them:

- **Member authentication:** existing email magic-link OTP flow.
- **Admin access control:** existing admin sidebar and role model. Electricity admin pages sit under the existing Member Services section of the admin sidebar.
- **Evidence vault anchoring:** existing evidence-vault service with SHA-256 content hashing. New event types: electricity_interval_batch_received, electricity_bill_reconciled, electricity_services_fee_issued, electricity_subsidy_paid, electricity_consent_event.
- **Godley accounting:** existing double-entry ledger with the new account keys listed in Part 9 added to the stewardship_accounts table.
- **Member Independence Vault (IV):** existing IV module; electricity services appear as a new tile in the Member IV home.

45. Evidence-vault anchoring events

45.1 Every material event in the programme is anchored to the evidence vault with a content hash and (on Hyperledger Besu migration) an on-chain commitment. The anchored event types are:

Event type	Content anchored
electricity_consent_event	Consent grant, revocation, or update — with timestamp and authenticated Member identifier.
electricity_interval_batch_received	SHA-256 of the Localvolts API response payload for each poll, with interval window and NMI scope.
electricity_bill_issued	SHA-256 of the Localvolts bill record as recorded in electricity_bills.
electricity_bill_reconciled	Reconciliation outcome hash; flags discrepancies between Localvolts-billed volumes and Foundation-held interval data.
electricity_services_fee_issued	Foundation's 15% COG$ invoice content hash.
electricity_subsidy_paid	STC-to-Localvolts AUD payment record hash.
electricity_class_c_offset_applied	Record of Class C offset applied against a services-fee invoice.
electricity_tier_change	Tier transition record with eligibility basis.

46. Build sequencing

46.1 Build sequencing follows the discipline of the existing platform. Each phase produces a single-commit deploy package.

Package	Contents
P1 — Schema	electricity_* table DDL; stewardship_accounts seeds for the new electricity account keys; migration SQL; no PHP changes. Ground-truth-clean proof precedes deploy.
P2 — LocalvoltsApiClient	PHP service class for Localvolts API integration; admin API-credentials page; ingestion cron; evidence anchoring of payloads. Member-facing

Package	Contents
	surfaces still hidden.
P3 — Member module, read-only	/services/electricity/index.php, dashboard.php, bills.php. Members can see their data but cannot onboard or change tier yet.
P4 — Member onboarding	/services/electricity/onboard.php full nine-step flow with Localvolts handoff and return. Tier management, consent capture.
P5 — Admin module	/admin/electricity/ full page set: cohort, intervals, reconciliation, subsidy processing, services-fee run, draw forecast, cases.
P6 — Optimiser engine	Fast, daily, and seasonal loops; Member dashboard live indicators; optimiser advice log.
P7 — Pilot go-live readiness	Monitoring, alerts, support runbooks, FNAC summary export, reconciliation dashboards. Final ground-truth proof and Phase 4 closure decision record.

PART 9 — GODLEY ACCOUNTING INTEGRATION

47. New account keys

47.1 The programme introduces new stewardship account keys following the established COMMUNITY-MOBILE pattern. These are added to the existing stewardship_accounts table and carry their own account_type values.

Account key	Purpose
COMMUNITY-ELECTRICITY-AR	Accounts receivable — 15% COG$ services fees invoiced to Members and not yet settled.
COMMUNITY-ELECTRICITY-REV	Revenue — 15% COG$ services fees settled (and if applicable, any referral acknowledgement from Localvolts flowing to Sub-Trust C).
COMMUNITY-ELECTRICITY-AP-SUPPORTED	Accounts payable — Sub-Trust C's outstanding obligations to Localvolts for Supported-tier Member AUD bills.
COMMUNITY-ELECTRICITY-SUBSIDY	Expense — Sub-Trust C AUD paid to Localvolts in settlement of Supported-tier bills.
COMMUNITY-ELECTRICITY-OFFSET-CLEARING	Clearing account — Class C COG$ offsets applied against 15% services-fee invoices. Net-zero at month end by design.

48. New Godley flows

48.1 Five new Godley flows are added. Each flow specifies sectors debited and credited, and the algebraic sum across sectors equals zero.

48.2 Flow E1 — Member electricity bill (Basic or Standard tier)

The Member pays Localvolts directly in AUD. The Foundation is not on the ledger for this flow. No Godley entry is recorded on the Foundation's books. The existence of the bill is recorded in electricity_bills for reference only.

48.3 Flow E2 — Foundation 15% services fee issued

Sector	Flow (COG$)	Explanation
MEMBER(n)	−X COG$	Services fee obligation created.
COMMUNITY-ELECTRICITY-AR	+X COG$	Foundation receivable increased.
ROW SUM	0	

48.4 Flow E3 — Services fee settled (cash)

Sector	Flow (COG$)	Explanation
MEMBER(n)	− (nil change in balance; offset of E2 obligation)	Obligation cleared by payment.
COMMUNITY-ELECTRICITY-AR	−X COG$	Foundation receivable reduced.
COMMUNITY-ELECTRICITY-REV	+X COG$	Revenue recognised.
ROW SUM	0	

48.5 Flow E4 — Services fee settled (Class C offset)

Sector	Flow (COG$)	Explanation
MEMBER(n)	− (Class C balance reduced)	Class C COG$ consumed as offset.
COMMUNITY-ELECTRICITY-AR	−X COG$	Foundation receivable cleared.
COMMUNITY-ELECTRICITY-OFFSET-CLEARING	+X / −X	Transient through clearing.
COMMUNITY-ELECTRICITY-REV	+X COG$	Revenue recognised.
ROW SUM	0	

48.6 Flow E5 — Supported-tier subsidy payment

Sector	Flow (AUD)	Explanation
STC-OPERATING	−Y AUD	Sub-Trust C cash outflow.
COMMUNITY-ELECTRICITY-SUBSIDY	+Y AUD	Subsidy expense recognised.
EXTERNAL-VENDOR (Localvolts)	+Y AUD	Counterparty paid on the Member's behalf.
COMMUNITY-ELECTRICITY-AP-SUPPORTED	−Y AUD	Prior accrued payable cleared.
ROW SUM	0	

49. Invariants inherited or extended

49.1 The programme inherits all existing Godley invariants from Specification v1.0. No invariants are relaxed. Two invariant-check view additions are specified:

- **EI1 — Services-fee reconciliation.** For each billing period, sum of COMMUNITY-ELECTRICITY-REV entries equals sum of 15% of AUD bill totals for Basic and Standard tier Members, less any waivers properly recorded in electricity_services_fees. Deviation > $0.01 triggers an exception.
- **EI2 — Sub-Trust C draw cap.** Aggregate month-to-date Sub-Trust C AUD outflow for electricity subsidies does not exceed the Board-approved monthly draw cap. Approaching the cap triggers a warning; breach triggers a suspension of new subsidy payments until Trustee decision record addresses the breach.

50. 30% First Nations allocation

50.1 Entrenched cl.27.3(d) requires that not less than 30% of annual Sub-Trust C distributions are directed to First Nations programs. The Foundation's existing invariant I10 monitors this ratio. COGS Community Electricity contributes to both sides of the ratio: (a) Sub-Trust C subsidy outflows supporting First Nations Members (including the Jubullum pilot cohort) count toward the 30% numerator; (b) 15% services-fee revenue flowing into Sub-Trust C counts toward the total distribution denominator and is available for First Nations allocation.

50.2 The Foundation monitors and reports the contribution of this programme to the 30% allocation as part of the FNAC periodic reporting cycle. The FNAC has visibility of both sides of the ratio as recorded in the Godley ledger.

PART 10 — CROSS-HUB GOVERNANCE

51. The five hubs

51.1 The Foundation's community projects are governed across five hubs. COGS Community Electricity is referenced in every hub. This section sets out the specific touchpoint and responsibility at each hub.

Hub	Touchpoint
First Nations	FNAC endorsement of programme scope; JLALC liaison on Jubullum cohort composition; FPIC baseline at supply-only phase; future Country-hosted generation FPIC at Phase 7+ when in scope.
Financial Oversight	Localvolts bill reconciliation; Sub-Trust C draw forecast and monthly reporting; GST treatment monitoring; 15% services-fee run review; Class C offset reporting; 30% First Nations allocation reconciliation.
Day-to-Day Operations	Member support runbook; life-support register handover; EWON escalation pathway; Localvolts partner relationship management; cohort monitoring and health reporting.
Technology and Blockchain	/services/electricity/ and /admin/electricity/ modules; Localvolts API integration; AI optimiser engine; evidence-vault anchoring of all programme events; Hyperledger Besu migration readiness.
Place-Based Decisions	Drake, Tabulam, Jubullum community briefings; cohort-level place-based reporting; distributor engagement; future community-generation siting discussions in Phase 7+.

52. Decision gates requiring cross-hub concurrence

52.1 Certain decisions require concurrence across multiple hubs before proceeding. These are:

- **Phase 0 closure.** Requires Caretaker Trustee decision with FNAC, Financial Oversight, and Technology hubs' material inputs.
- **Pilot cohort composition.** Requires FNAC concurrence; Place-Based Decisions hub sign-off for Drake/Tabulam/Jubullum ordering.
- **Class C offset cap values.** Requires Board (or Caretaker Trustee during Caretaker period) decision with Financial Oversight input and FNAC notice.
- **Services-fee percentage change.** Requires Board decision with FNAC concurrence where the Community Supported tier is affected.
- **Retailer partner change.** Requires Caretaker Trustee decision; material input from Financial Oversight, Technology, and Day-to-Day Operations hubs; FNAC notice.

53. Reporting cadence

53.1 The programme's standing reporting cadence is:

- **Monthly:** Financial Oversight report on 15% services-fee income, Sub-Trust C subsidy outflow, Class C offset usage, reconciliation status.
- **Quarterly:** Member survey on AI optimisation usefulness; FNAC report including 30% First Nations allocation contribution; Technology-hub report on optimiser quality metrics.
- **At 3 months and 6 months post-soft-launch:** Formal pilot review against §40 success conditions.
- **Annually:** Full programme review by the Caretaker Trustee / Board; FNAC annual review; services-fee percentage review; cohort-expansion planning.

PART 11 — SEVEN-PHASE TIMELINE

54. Overview

54.1 COGS Community Electricity is a seven-phase programme, designed to run in parallel with COGS Community Mobile where possible and to stagger where electricity-specific dependencies (GST, life-support, distributor engagement) require sequential steps. Dates are indicative and subject to Phase 0 closure.

55. Phase 0 — Readiness (May–July 2026)

55.1 Activities: Trustee decision record; FNAC/JLALC concept engagement; GST and tax advice (including any private-ruling application); legal review of NERL, MIS, and fee characterisation; AML/CTF refresh; distributor ground-truth check; community information briefings; Localvolts concept note and exploratory call; life-support handover protocol; Phase 0 closure decision record.

55.2 Gate: Phase 0 closure decision record signed. No Phase 1 activity begins until this is on file.

56. Phase 1 — Commercial partnership (July–August 2026)

56.1 Activities: Localvolts Partnership Arrangement drafted, reviewed, and executed; Partner API key and Partner ID issued; co-branding parameters agreed; onboarding-channel specifics agreed; Foundation-side AML/CTF and data-handling artefacts aligned with the Partnership Arrangement.

56.2 Gate: executed Partnership Arrangement and working API credentials in the Foundation's admin credential store.

57. Phase 2 — Platform build (August–October 2026)

57.1 Activities: build packages P1 through P5 (see §46). Schema, LocalvoltsApiClient, Member module, onboarding flow, admin module. Godley accounting integration. Evidence-vault anchoring for all defined event types. End-to-end test across non-pilot test accounts.

57.2 Gate: end-to-end test on test-account NMIs produces correct bills, correct services-fee invoices, correct evidence anchors, clean Godley reconciliation. Ground-truth proof clean.

58. Phase 3 — Optimiser go-live (September–October 2026)

58.1 Activities: build package P6. Optimiser engine fast, daily, and seasonal loops. Dashboard live tiles. Advice logging. Quality measurement harness. Life-support exclusion check. Advisory-only; no device control.

58.2 Gate: optimiser quality metrics within expected bounds on historical data replay; dashboard live in staging; operations runbook complete.

59. Phase 4 — Soft launch (October–November 2026)

59.1 Activities: pilot cohort onboarding (30–60 Member premises across Jubullum, Drake, Tabulam). First Localvolts bills. First 15% COG$ services-fee run. First Sub-Trust C subsidy payments. First cohort support cases. Monitoring and alerting actively watched.

59.2 Gate: 6-month pilot review (April–2027) against §40 success conditions.

60. Phase 5 — NSW general availability (Q2 2027)

60.1 Activities: onboarding opens to any NSW Member. Essential Energy, Ausgrid, and Endeavour distribution areas all in scope. Cohort scale moves from tens to hundreds of premises. Optimiser moves to real-time 5-minute signals for Members with compatible smart devices. Monthly published programme report.

60.2 Gate: six-month steady-state operation at NSW-wide cohort.

61. Phase 6 — NEM-wide (ex-VIC) (late 2027)

61.1 Activities: extend onboarding to ACT, QLD, SA, TAS via Localvolts' existing coverage. Jurisdictional overlays reviewed for each state. Each new state's life-support register and EWON equivalent confirmed. Distributor engagement extended.

61.2 Gate: operational viability confirmed in each new jurisdiction before the next is opened.

62. Phase 7 — Victoria and JV generation (2028 and beyond)

62.1 Activities: second retailer partner engaged for Victoria (Amber Electric, Next Business Energy, or Diamond Energy as surveyed in §18.3). Victorian licence-specific issues addressed. Separately, planning begins for community-scale generation projects (solar, battery, microgrid) where Country hosting and FPIC preconditions are met. Each such project has its own FNAC engagement and its own founding planning document.

62.2 Gate: (a) Victoria onboarding live; (b) JV generation projects each gated by their own founding documents, not by this document.

Parallel operation with Community Mobile

Wherever possible, Electricity and Mobile phases run in parallel. Phase 0 activities (Trustee decision, FNAC engagement, GST advice, legal review, AML/CTF, community briefings, concept notes to partners) can be sequenced so the same community briefings cover both programmes. Phase 2 platform build reuses existing /services/ and /admin/ patterns. Phase 4 soft launches can target overlapping Member cohorts, reducing cognitive load on pilot Members.

PART 12 — RISK REGISTER

63. Risk posture

63.1 The Foundation's risk posture for COGS Community Electricity is deliberately conservative. Electricity is a critical service, regulated environment, and high-cost item for Members; Member trust in the programme and in the Foundation must not be traded against speed of launch.

63.2 Risks are grouped into five categories: regulatory, commercial, operational, First Nations governance, and Member financial. Each risk has a description, a likelihood assessment (L/M/H), an impact assessment (L/M/H), and a primary mitigation.

64. Risk register

Risk	Description	Lik.	Imp.	Primary mitigation
R1 NERL reclassification	Regulator or a party argues the 15% fee is a retail supply consideration bringing the Foundation within NERL.	L	H	Phase 0 legal opinion on fee characterisation; fee is for services, not supply; no Foundation holding of title to electricity.
R2 MIS-posture drift	The combined electricity-plus-COG$ arrangement is read as disturbing the existing MIS-exclusion posture.	L	H	Phase 0 legal review specifically on MIS-posture consistency; programme structure is Member↔retailer bilateral, not pooled.
R3 GST recharacterisation	ATO takes a view that GST is chargeable on the 15% COG$ fee despite the Sub-Trust C flow-through framing.	M	M	Phase 0 tax advice or private-ruling application before launch; if GST is chargeable, programme pricing and Sub-Trust C posture adjusted before launch.
R4 Localvolts partnership decline	Localvolts declines or cannot engage on the requested terms.	M	M	Prepared fallback to Next Business Energy Pty Ltd, Amber Electric, or Diamond Energy

Risk	Description	Lik.	Imp.	Primary mitigation
				under §18.3.
R5 Localvolts RoLR event	Localvolts is suspended from the NEM and Members transfer to a default RoLR.	L	M	Members are protected by the AER RoLR scheme; Foundation pauses services-fee invoicing; accelerated rebrand onto a replacement retail host.
R6 Wholesale price spike	A severe wholesale price event increases Supported-tier Sub-Trust C outflow beyond budget.	M	M	Monthly Sub-Trust C draw cap (EI2); escalation to Trustee decision record at 80% of cap; Member advice during known spike windows.
R7 Class C offset exhaustion	Members unintentionally exhaust Class C balances during a high-usage period.	M	L	Per-Member monthly and rolling 12-month caps; dashboard warning at 70% of cap.
R8 Life-support incident	Optimisation advice is misinterpreted as a reason to reduce supply at a life-support premises.	L	H	Explicit life-support declaration at onboarding; life-support exclusion filter in optimiser; advice copy specifically worded to preserve supply in all edge cases; no device control in Phase 1.
R9 Data breach	Member interval data or identifying info is exposed through Foundation platform or transit path.	L	H	Existing platform data-protection controls; data minimisation; evidence-vault content hashing; TLS transport; API credentials in server-side secret

Risk	Description	Lik.	Imp.	Primary mitigation
				store.
R10 First Nations governance breakdown	FNAC or JLALC withdraws concurrence.	L	H	Early and continuous engagement; explicit opt-out at Phase 0 item 7; staged onboarding respects JLALC timing; no cohort progression without FNAC concurrence.
R11 API version break	Localvolts API v1 introduces a breaking change.	L	L	Version-probe on /v1/version endpoint; staged rollout across test accounts before production; close partnership with Localvolts for early notice.
R12 Optimiser advice drift	Optimiser recommendations routinely miss the actual cheapest window.	M	L	Advice-quality measurement harness; threshold for triggering engineering review; rule-based Phase 1 design keeps the model auditable.
R13 COG$ barter misunderstanding	Members treat COG$ services-fee invoices as conventional currency and under-plan payment.	M	L	Clear onboarding disclosure; dashboard balance indicators; 60-day grace before any suspension of services for unpaid COG$ fees; waiver pathway always available.
R14 Member-Localvolts friction	A Member raises a dispute that blurs lines between retail issue (Localvolts) and community services issue (Foundation).	M	L	Clear pathway disclosure at onboarding; Foundation support team trained to triage and redirect;

Risk	Description	Lik.	Imp.	Primary mitigation
				EWON escalation supported on the retail side.
R15 Foundation operational overload	Programme scale exceeds operational capacity during Phase 5 NSW general availability.	M	M	Phased expansion by distribution area; monthly health reporting; pause criteria documented; hiring/contract resourcing aligned to cohort scale.

64.1 This register is reviewed quarterly by the Caretaker Trustee and updated as risks change. New risks identified in pilot operation are added with the same discipline.

PART 13 — READINESS STATEMENT AND SIGNATURES

65. Specification completeness

65.1 This document, together with:

- the COGS Community Mobile Planning and Strategy Document (template and structural reference);
- the Joint Venture Participation Agreement and CJVM Hybrid Trust Declaration (supreme instruments);
- the Sub-Trust C Deed (remediated R6) governing subsidy funding;
- the Godley Accounting Specification v1.0 (post-counsel review);
- the Localvolts API Guide v0.9.2 (authoritative for the integration);
- the Localvolts Partnership Concept Note v1.0 (issued as the first commercial document); and
- the Foundation's current platform source at repo cogsaustralia/Cog,

— constitutes a complete specification for Phase 0 execution and Phase 1 commercial engagement.

66. Confirmation items

66.1 Three items require confirmation before Phase 0 commences in earnest:

- **Item A.** The Phase 0 checklist ordering and gating in §35 is acceptable to the Caretaker Trustee as the operative sequence.
- **Item B.** The 15% services-fee characterisation as a retail services margin and infrastructure-testing nominal value, with GST flow-through via Sub-Trust C, is the characterisation the Foundation will present to its regulatory counsel and tax adviser.
- **Item C.** Localvolts is the Phase 1 first-choice retail partner and the fallback sequence (Next Business Energy → Amber → Diamond) is the fallback order.

66.2 On confirmation of these three items, Phase 0 item 1 (Trustee decision record) can be executed and the programme formally commences.

67. Amendment history

Version	Date	Change
1.0	25 April 2026	Initial issue as COGS Community Electricity. Scope: supply-only retail through Localvolts; Jubullum/Drake/Tabulam pilot; 15% COG$ services fee; GST flow-through via Sub-Trust C; AI optimisation layer; seven-phase timeline through JV generation at Phase 7+.
1.1	25 April 2026	Structural framing revision aligning the programme with the COGS Resource Corridors (CRC) umbrella. Programme renamed to Corridors Energy: Electricity. New §1A placing the programme in the CRC Energy branch and referencing the CRC Master Strategy v1.0

Version	Date	Change
		for the common substrate (Member identity, COG$ mechanics, Sub-Trust C three-tier, Godley ledger, evidence vault, FNAC governance, Phase 0 discipline). Engineering identifiers adopt the CORRIDORS-* convention for new accounts; existing COMMUNITY-ELECTRICITY-* keys retained for continuity. No substantive change to commercial model, regulatory posture, timeline, or Phase 0 checklist.

68. Signature block

Thomas Boyd Cunliffe

Declarant and Caretaker Trustee

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